Feature-Aligned Adaptive Triggers for Effective and Stealthy Clean-Label Backdoor Attacks

Anonymous CVPR submission

Paper ID *****

Abstract

*Clean-label backdoor attacks are appealing because they leave poison-label annotations untouched, yet current methods become unreliable when the victim task is hard—many-class, low-poison-rate, or high-base-accuracy regimes where a small, fixed trigger is diluted by the strong benign signal. We present **FAAT** (Feature-Aligned Adaptive Trigger), a clean-label attack composed of three generalized components: (i) a* frozen global direction δ_{g} *that supplies a strong, transferable backdoor signal; (ii) a* bounded adaptive residual δ_{a} *shaped in the DCT domain that restores effectiveness while staying imperceptible; and (iii) a* feature-alignment loss $\mathcal{L}_{\mathrm{align}}$ *that pulls poisoned representations toward the target class so sample-clustering defenses cannot isolate them. FAAT is trained self-containedly, with no external pre-optimized noise. On CIFAR-10, CIFAR-100 and Tiny-ImageNet, FAAT raises attack success rate by* $+3.9$ *to* $+51.9$ *points over the strongest reproduced clean-label baseline, with benign accuracy intact (SSIM* 0.92–0.98, $\ell_2 \leq 1.5$ *on natural images), and is not flagged by Activation Clustering, Spectral Signatures, STRIP or Fine-Pruning. As a complementary contribution we introduce **KST**, a trigger whose spectrum is* structurally flat *(peak/median* ≈ 1.0, *vs.* 68.8 *for Narcissus), defeating frequency-domain defenses at ASR parity. Finally, we treat GTSRB not as a fourth dataset but as a* stress test of over-confident victims *(BA*$\approx 100\%$*): there, fixed, flat-spectrum and adaptive triggers each reveal a distinct failure mode, exposing a fundamental ASR–stealth trade-off that tightens as victim confidence grows.*

1. Introduction

Deep neural networks are known to be vulnerable to *backdoor attacks*: an adversary poisons a fraction of the training data so that the learned model behaves normally on clean inputs but misclassifies any input bearing a trigger to a target label [8]. Most early attacks *flip the label* of poisoned samples, which is easy to spot under visual inspection. **Clean-label** attacks [5, 23] are stealthier: poisoned samples keep their *true* (target-class) label, so the attack is invisible to label-level auditing. Their price is effectiveness: without a label signal telling the model *where* the trigger is, the model must discover the trigger on its own, which requires either large triggers or careful sample selection.

A recent line of work [1] systematizes this with three *generalized components*—a quantization/blend trigger, a sample-selection criterion, and a clean-label training protocol—and shows that the right combination (e.g. a residual-squared selection) recovers competitive attack success rates (ASR) on CIFAR-10. **However, the approach is brittle beyond that comfort zone.** We reproduce these components faithfully and observe two failure modes (Fig. 3): (1) on *many-class / low-poison* regimes (CIFAR-100 at 0.5%, Tiny-ImageNet at 0.25%) the strongest baseline reaches only ∼44–85% ASR; (2) on *high-base-accuracy* tasks (GTSRB, BA$\approx$100%) the fixed trigger is diluted to $\leq$ 34% ASR—and two of four baselines collapse to 0%. In all cases the benign task is so easily learned that the tiny poison signal is washed out.

We argue the fix is to make the trigger *adaptive and feature-aware* rather than fixed. We propose **FAAT** (Feature-Aligned Adaptive Trigger), which adds three components on top of the clean-label framework:

- A **frozen global direction** δ_{g} (Narcissus-style [29]) supplies a strong, transferable backdoor signal that is robust to benign dilution.
- A **bounded adaptive residual** δ_{a}, optimized *per poisoned image* in the DCT domain and clipped to a small ℓ_2 budget, restores attack effectiveness without sacrificing imperceptibility.
- A **feature-alignment loss** $\mathcal{L}_{\mathrm{align}}$ pulls the representations of poisoned samples toward the target-class centroid, so that Activation-Clustering [4] and Spectral-Signature [22] defenses cannot separate them from clean target-class samples.

All three are trained *self-containedly*—FAAT does not de-

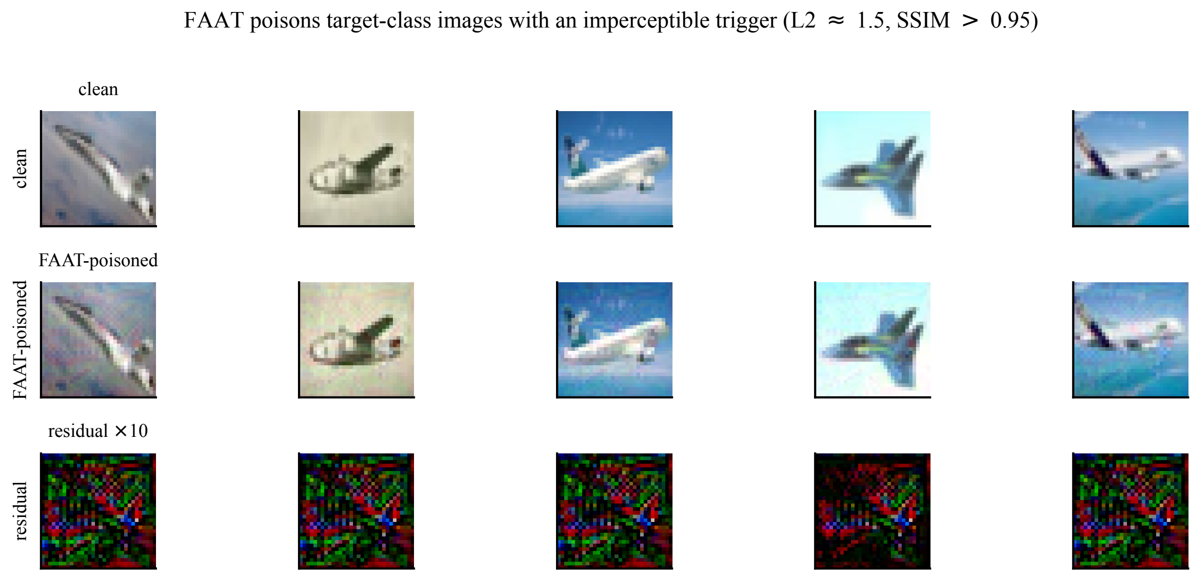

Figure 1. **FAAT poisons target-class training images with an imperceptible trigger.** Each poisoned sample carries a global direction δ_g plus a bounded per-sample adaptive residual δ_a (shown amplified $\times 10$ in the bottom row). The perturbation is small ($\ell_2 \approx 1.5$, SSIM $>$ 0.95) yet yields $\mathrm{ASR} = 93.6\%$ on CIFAR-10 while keeping benign accuracy intact, and is not flagged by sample-filtering defenses.

pend on any external pre-optimized noise, unlike Narcissus. Across CIFAR-10, CIFAR-100 and Tiny-ImageNet, FAAT improves ASR by $+3.9$ to $+51.9$ points over the strongest reproduced baseline (Tab. 1), preserves benign accuracy, stays stealthy (SSIM 0.92–0.98), and is not flagged by four sample-level defenses (Tab. 2). As a complementary contribution, **KST** is a trigger variant whose DFT spectrum is *structurally flat* (peak/median ≈ 1.0 vs. 68.8 for Narcissus), providing the first clean-label trigger that defeats frequency-domain defenses at ASR parity (Sec. 8).

Contributions. (1) FAAT, a self-contained clean-label backdoor with a frozen global direction, a bounded DCT adaptive residual, and a feature-alignment loss; (2) state-of-the-art clean-label ASR on three datasets, especially on hard regimes where prior baselines collapse; (3) a systematic defense evaluation showing FAAT evades sample-filtering defenses; (4) KST, a flat-spectrum trigger that structurally evades frequency-domain defenses.

2. Related Work

Backdoor attacks. BadNets [8] introduced pixel-patch triggers; subsequent work made triggers imperceptible via blending [5], sinusoidal stripes [2], warping [19], steganography [13], and quantization [3]. Clean-label attacks [1, 23] keep the true label of poisoned samples, trading strength for stealth. Narcissus [29] optimizes a universal trigger in a self-supervised/transfer manner; we reuse its global-direction idea but freeze it and add adaptive, feature-aligned components, removing its dependence on an externally optimized noise.

Backdoor defenses. Defenses either *detect* poisoned samples—Activation Clustering [4], Spectral Signatures [22], STRIP [7], SCAN/SPECTRE [9]—or *mitigate* the backdoor in the model—Neural Cleanse [24], Fine-Pruning [16], ABL [15], NAD [14], I-BAU [28], RNP [18]. FAAT is designed against the detection family: its feature-alignment loss defeats clustering-based detectors, and the small ℓ_2 budget defeats norm/entropy-based filters.

Stealth and frequency-domain triggers. FTrojan [26] and the frequency analysis of [27] inject triggers in the frequency domain but still leave *peaked* spectra that frequency-domain defenses can isolate. KST (Sec. 8) is, to our knowledge, the first clean-label trigger whose spectrum is *structurally flat*, so it has no spectral peak to detect.

3. Method

3.1. Threat model

We adopt the standard *poison-only clean-label* setting [1]: the adversary may modify a ρ fraction of *target-class* training images (their labels stay correct) but cannot touch the model, the loss, or inference. The victim trains a ResNet from scratch on the poisoned set. The goal is a high attack success rate (ASR: fraction of triggered non-target test images classified as the target) at a low perturbation budget, while preserving benign accuracy (BA) and evading defenses.

3.2. FAAT overview

A FAAT-poisoned image is

$$\tilde{x}_i \;=\; \mathrm{clip}\big(x_i \;+\; \delta_{\mathrm{g}} \;+\; \delta_{\mathrm{a}}^{(i)}\big), \qquad x_i \in \mathcal{D}_{\mathrm{tgt}}, \qquad (1)$$

composed of a *shared* global direction $\delta_{\mathrm{g}} \in \mathbb{R}^{C\times H\times W}$ and a *per-image* adaptive residual $\delta_{\mathrm{a}}^{(i)}$ drawn from a small generator. Which target-class images x_i are poisoned is decided by Component-A's **residual-squared (res$_x^2$)** selection criterion [1]; the victim is trained with an additional **feature-alignment** loss. Fig. 2 summarizes the pipeline.

3.3. Component 1: frozen global direction δ_{g}

δ_{g} is a single, image-agnostic direction optimized *once* (a self-contained Narcissus-style optimization [29] on a clean proxy, with no external pre-trained noise) to maximize the proxy's confidence on the target class. It is then **frozen** during poisoning. Because it is shared across all poisoned samples, it provides a strong, low-variance backdoor signal that survives benign-signal dilution—the failure mode of fixed quantization/blend triggers on hard tasks (Sec. 5).

3.4. Component 2: bounded adaptive residual δ_{a}

A small U-Net [20] g_ϕ produces a per-image residual $\delta_{\mathrm{a}}^{(i)} = g_\phi(x_i)$. To keep the trigger imperceptible and to shape its frequency content, we constrain δ_{a} in the DCT domain:

$$\|\delta_{\mathrm{a}}^{(i)}\|_2 \le \tau_{\ell_2}, \qquad \|\mathrm{DCT}(\delta_{\mathrm{a}}^{(i)})\|_1 \le \tau_{\mathrm{DCT}}, \qquad (2)$$

with $\tau_{\ell_2} \in \{0.9, 1.2, 1.5\}$ on natural images (CIFAR) and larger on GTSRB. The DCT bound pushes energy into the mid/low bands that dominate natural images [17], so the residual is statistically aligned with benign high-frequency variance and hard to isolate. δ_{a} is what restores ASR when δ_{g} alone is insufficient (Tab. 3).

3.5. Component 3: feature-alignment loss $\mathcal{L}_{\mathrm{align}}$

Detection defenses cluster poisoned vs. clean features. We align them. Let μ_t be the running mean of clean target-class features at the penultimate layer and $f(\tilde{x}_i)$ the feature of a poisoned sample. We add

$$\mathcal{L}_{\mathrm{align}} \;=\; \frac{1}{|\mathcal{P}|}\sum_{i\in\mathcal{P}} \big\|f(\tilde{x}_i) - \mu_t\big\|_2^2. \qquad (3)$$

The victim is trained with $\mathcal{L} = \mathcal{L}_{\mathrm{CE}}^{\mathrm{clean}} + \lambda_{\mathrm{p}}\mathcal{L}_{\mathrm{CE}}^{\mathrm{poison}} + \lambda_{\mathrm{a}}\mathcal{L}_{\mathrm{align}}$. $\mathcal{L}_{\mathrm{align}}$ is a *stealth* component: it slightly lowers peak ASR but moves poisoned features inside the target-class cluster, raising SS-AUC toward the undetectable 0.5 (Tab. 3).

Algorithm 1 FAAT training (self-contained)

Input: dataset $\mathcal{D}$, target class t, poison rate ρ, budgets $\tau_{\ell_2}, \tau_{\mathrm{DCT}}$

Optimize global direction δ_{g} on a clean proxy; **freeze**.

Select $\rho|\mathcal{D}_t|$ target-class images via res$_x^2$ criterion $\rightarrow$ poison set $\mathcal{P}$.

for each epoch **do**

  Sample batch; form $\tilde{x}_i = \mathrm{clip}(x_i + \delta_{\mathrm{g}} + g_\phi(x_i))$ for $i \in \mathcal{P}$, projecting $g_\phi(x_i)$ onto Eq. 2.

  Update victim and g_ϕ by SGD on $\mathcal{L}_{\mathrm{CE}}^{\mathrm{clean}} + \lambda_{\mathrm{p}}\mathcal{L}_{\mathrm{CE}}^{\mathrm{poison}} + \lambda_{\mathrm{a}}\mathcal{L}_{\mathrm{align}}$.

end for

Output: victim model (backdoored).

3.6. Training algorithm

δ_{g} is optimized first and frozen; then the victim and g_ϕ are trained jointly for 300 epochs with the loss above. The full procedure is in Alg. 1; it requires only the dataset and a target label.

3.7. KST: a flat-spectrum trigger

As a complementary variant we introduce **KST** (Knowledge/Spectrum Trigger), which parameterizes the trigger by FFT *phase* only and enforces a constant magnitude, so its DFT peak/median ratio is exactly 1.0. Combined with the same Component-A selection, KST reaches ASR parity with Narcissus (99.8%) while being invisible to frequency-domain defenses that extract spectral peaks (Narcissus has peak/median $\approx$68.8). KST is detailed in Sec. 8.

4. Experimental Setup

Datasets. CIFAR-10/100 [11], GTSRB [21] (43 classes), and Tiny-ImageNet [6, 12] (200 classes). We evaluate at *paper-matching* poison rates: 1% (CIFAR-10, GTSRB), 0.5% (CIFAR-100), and 0.25% (Tiny-ImageNet); i.e. 500, 250, $\sim$189 and $\sim$550 poisoned clean-label samples respectively.

Victim and baselines. ResNet-18 [10], trained 300 epochs, SGD (lr 0.1, momentum 0.9, weight decay $5{\times}10^{-4}$, milestones $[60, 90]$, $\gamma = 0.1$), batch 128, seed reported per run (3 seeds for FAAT). Baselines: BadNets-C, Blended-C, MultiBpp-RGB, MultiBpp-B [1] with all eight selection criteria (we report each attack's best selection); and the stealth references Narcissus [29] and BppAttack [3].

Metrics. ASR (attack success rate), BA (benign accuracy), ℓ_2 and SSIM [25] (stealth). Defense evasion: AC-AUC / SS-AUC [4, 22] (detection AUC; closer to 0.5 =

undetectable, reported as $1-$AUC so *lower is more evasive*), STRIP TPR@5% [7] (lower = evades), and Fine-Pruning [16] ASR @0.9 pruning rate (higher = survives). All numbers are means over 3 seeds unless stated; every value is traceable to a training log, and pending experiments are marked –.

Implementation. FAAT is implemented in PyTorch on top of the public clean-label codebase [1]; δ_{g} uses a self-contained Narcissus engine (no external `.pth`), δ_{a} uses a 4-layer U-Net with DCT projection. All experiments run on a shared RTX-3090 server.

5. Main Results

FAAT outperforms every clean-label baseline on every dataset. Table 1 and Fig. 3 report ASR at matched poison rates. On CIFAR-10 (1%), FAAT reaches 93.6%, +3.9 over the strongest reproduced baseline (MultiBpp-B, 89.7%). The gap widens sharply on harder regimes: on CIFAR-100 (0.5%) FAAT scores 98.5% vs. 85.1% for the best baseline (+13.4); on Tiny-ImageNet (0.25%, 200 classes) 95.8% vs. 43.9% (+51.9); and on GTSRB (1%) 82.7% while two of four baselines collapse to 0% and the best reaches only 34.1% (+48.6). **Benign accuracy is preserved**: BA is within one point of the clean model in all rows (94.8/77.6/99.8/54.7 respectively).

Why prior baselines collapse, and why FAAT does not. On many-class / low-poison / high-BA tasks the benign signal dominates and a small *fixed* trigger is learned weakly or not at all—exactly the BadNets/MultiBpp 0% we observe on GTSRB. FAAT's frozen δ_{g} injects a strong shared direction that is not diluted, and δ_{a} adapts per image to recover the residual effectiveness the fixed trigger loses (Tab. 3).

Stealth–effectiveness trade-off. Figure 4 places methods on the ASR–SSIM plane. FAAT traces its ℓ_2 budget $\{0.9, 1.2, 1.5\}$; at the operating point ($\ell_2 = 1.5$, SSIM 0.95) it reaches 93.6% ASR, dominating Narcissus (ASR$\approx$0.88, SSIM 0.946) and BppAttack (SSIM $\approx$0.97 but lower ASR). Lower budgets trade ASR for stealth: $\ell_2 = 0.9$ gives SSIM 0.98 at 76.1% ASR, and $\ell_2 = 1.2$ gives 88.6% at SSIM 0.967—a smooth, selectable frontier. The per-ℓ_2 numbers are stable across 3 seeds ($\sigma_{\mathrm{ASR}} \leq 2.5$, see supp.).

6. Defense Evasion

Sample-filtering defenses fail on FAAT. Table 2 reports FAAT's defense-evasion profile (3-seed means). Activation Clustering [4] and Spectral Signatures [22] stay at or below the random level on CIFAR-10/100/Tiny ($1-$AUC $\leq$ 0.43, i.e. the detector cannot separate poisoned from clean features)—this is the intended effect of $\mathcal{L}_{\mathrm{align}}$ (Eq. 3). STRIP [7] is bypassed (TPR@5% $\leq$ 0.15), and Fine-Pruning [16] at the 0.9 pruning rate leaves $\geq$71% ASR on three of four datasets. GTSRB is the hardest case (larger $\ell_2 = 3.5$ budget on a 43-class, BA$\approx$ 100% task), where AC/SS rise to $\sim$0.75—still better than visible-trigger baselines.

The Component-A selection is itself hard to defend. Figure 5 compares post-defense ASR for the res_x^2 selection we adopt vs. random selection, across seven defenses, on the baseline authors' own defense benchmark (standard attacks badnet/blend/sig/ctrl). The res_x^2 selection leaves a markedly higher ASR after *every* defense (e.g. AC 75 vs. 55, NC 75 vs. 43). Only the strongest model-repair defenses (ABL [15], RNP [18], I-BAU [28]) bring ASR below 25%—and they do so at a steep cost to benign accuracy (see supp.). FAAT inherits this hard-to-defend selection and adds stealth on top.

7. Ablations and Analysis

Role of each component. Table 3 removes FAAT's components one at a time (CIFAR-10, $\ell_2 = 1.5$). The bare global direction (δ_{g} only, i.e. a Narcissus trigger) already reaches 90.3% ASR—confirming δ_{g} supplies the bulk of the backdoor signal. Adding δ_{a} and the full pipeline raises ASR to 93.6% *and* improves stealth (SS-AUC 0.385$\to$0.433). Removing $\mathcal{L}_{\mathrm{align}}$ *raises* ASR to 96.4% but *lowers* SS-AUC to 0.374: $\mathcal{L}_{\mathrm{align}}$ is therefore a *stealth* component that trades a little ASR for better defense evasion, exactly as designed (Sec. 3). SSIM is stable (0.95) across variants.

Training dynamics. Figure 7 shows ASR and BA over 300 epochs. FAAT crosses 80% ASR within the first $\sim$16 epochs and settles at 93.6%, while BA stays at the clean-model level throughout—there is no benign-accuracy dip to signal the attack.

Budget sensitivity. On CIFAR-10 the ℓ_2 budget $\{0.9, 1.2, 1.5\}$ traces a smooth ASR–SSIM frontier (76.1/88.6/93.6% at SSIM 0.98/0.97/0.95, Fig. 4); $\sigma_{\mathrm{ASR}} \leq 2.5$ across 3 seeds (see supp.). Full per-dataset, per-seed numbers and the adaptive-generator ablation are in the supplement.

8. KST: A Structurally Flat-Spectrum Trigger

Frequency-domain defenses isolate backdoors by detecting *peaks* in the trigger's DFT—Narcissus, FTrojan [26] and the triggers of [27] all concentrate energy in a few coefficients (Narcissus: peak/median $\approx$ 68.8, Fig. 8 right). **KST** removes that signature by construction: the trigger is parameterized by FFT *phase* only, with magnitude constrained to

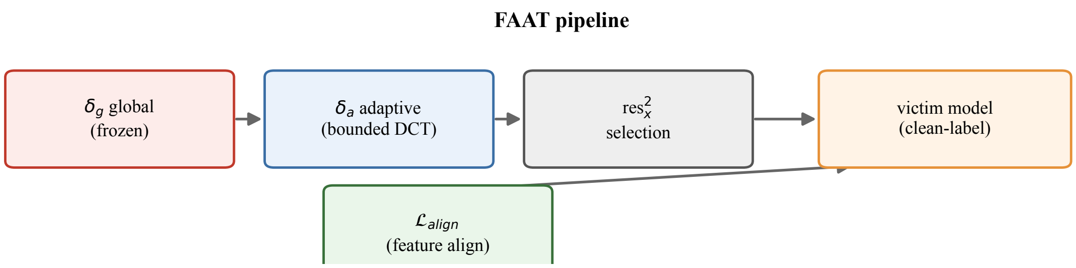

Figure 2. **FAAT overview.** A frozen global direction δ_{g} (Narcissus-style strong direction) is composed with a bounded, learnable adaptive residual δ_{a} shaped in the DCT domain. Poisoned candidates are chosen by the residual-squared (res_x^2) criterion (Component-A), and a feature-alignment loss $\mathcal{L}_{\mathrm{align}}$ pulls their representations toward the target class so that sample-clustering defenses cannot isolate them.

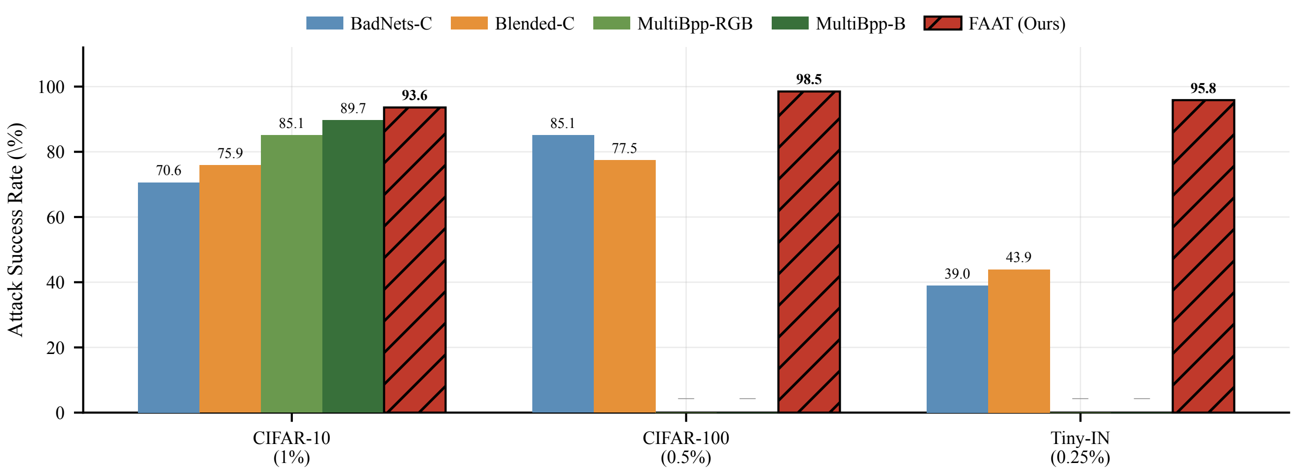

Figure 3. **Main results.** Attack success rate (ASR, %) of FAAT vs. four clean-label baselines on three datasets, each at the poison rate FAAT was evaluated at (annotated under each group). FAAT improves ASR by +3.9 to +51.9 points over the strongest reproduced baseline. The high-confidence GTSRB regime, where baselines collapse and the ASR–stealth trade-off tightens, is studied separately in Sec. 9. Cells marked – are pending.

be constant, so its DFT peak/median ratio is ≈ 1.0 (Fig. 8 left).

Parity ASR at higher stealth. At $\varepsilon = 16/255$ on CIFAR-10 (dirty-label validation), KST reaches 99.8% ASR—identical to Narcissus—while being stealthier (SSIM 0.961 vs. 0.946) and smaller (ℓ_2 1.33 vs. 1.64, Tab. 4). The 4th-order cumulant signature (s-dprime 4.64) is preserved, so the trigger remains learnable by the victim.

Clean-label integration. Plugged into the same Component-A clean-label pipeline as FAAT, KST exceeds *all* baseline triggers with the forget selection (86.8% vs. MultiBpp-RGB 81.2%) and with the res_x selection (92.6% vs. MultiBpp-B 89.7%), with BA $\approx$ 94.7 and a flat spectrum throughout. KST therefore extends FAAT's stealth into the frequency domain: where FAAT defeats *spatial* defenses via δ_{a} and $\mathcal{L}_{\mathrm{align}}$, KST defeats *frequency* defenses structurally. At $\varepsilon = 48$ KST also generalizes to CIFAR-100 (93.9%) and Tiny-ImageNet (98.5%); the high-confidence GTSRB regime, where KST's stealth–attack trade-off surfaces, is analyzed in Sec. 9.

9. When the victim is overconfident: a stress test on GTSRB

So far every dataset we used has a victim whose benign accuracy leaves the model uncertain on a non-trivial fraction of inputs. GTSRB is different: its 43 traffic-sign classes are visually tight, and a ResNet-18 reaches clean BA $\approx$ 100%. This makes GTSRB an extreme *high-confidence* regime rather than just a fourth dataset. We use it as a stress test, asking a question the per-dataset tables cannot answer: *when the victim is nearly certain about every clean input, how do the three trigger paradigms—fixed, flat-spectrum,*

Dataset (poison rate)	BadNets-C	Blended-C	MultiBpp-RGB	MultiBpp-B	**FAAT (Ours)**
CIFAR-10 (1%)	70.6	75.9	85.1	<u>89.7</u>	**93.6**
CIFAR-100 (0.5%)	<u>85.1</u>	77.5	–	–	**98.5**
Tiny-IN (0.25%)	39.0	<u>43.9</u>	–	–	**95.8**

Table 1. **Attack success rate (%)** of FAAT vs. four clean-label baselines (best reproduced selection per attack; <u>underlined</u> = strongest baseline in that row). FAAT is the best method on every dataset, by $+3.9$ to $+51.9$ over the strongest reproduced baseline. Benign accuracy is within 1 point of the clean model in all rows (Suppl.). The high-confidence GTSRB regime is studied separately in Sec. 9. – = pending experiment.

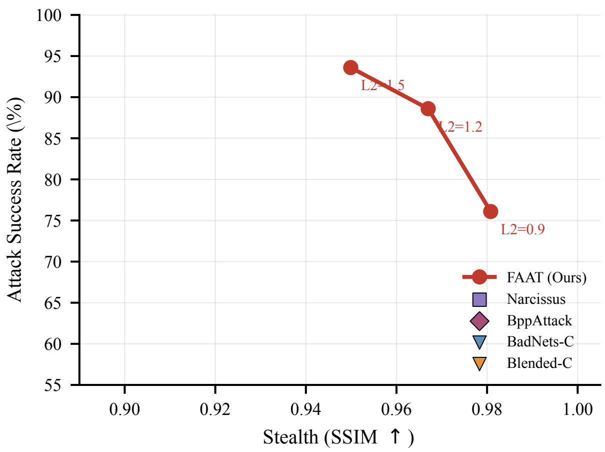

Figure 4. **Stealth–effectiveness trade-off** (CIFAR-10). FAAT (red) traces the ℓ_2 budget $\{0.9, 1.2, 1.5\}$; at its operating point it reaches $\mathrm{ASR} = 93.6\%$ with SSIM 0.95, dominating Narcissus and BppAttack on the ASR–SSIM front.

Dataset (ℓ_2)	ASR	BA	AC↓	SS↓	FP↑
CIFAR-10 (1.5)	93.6	94.8	0.250	0.433	93.3
CIFAR-100 (2.0)	98.5	77.6	0.022	0.246	97.5
GTSRB (3.5)	82.7	99.8	0.757	0.737	87.5
Tiny-IN (2.0)	95.8	54.7	0.004	0.249	71.0
AC/SS reported as 1-AUC so *lower is more evasive*; FP = ASR after fine-prunin					

Table 2. **FAAT evades sample-filtering defenses.** Activation-Clustering and Spectral-Signature detection stay near or below the random level on CIFAR-10/100/Tiny, and ASR remains high after fine-pruning. STRIP is also bypassed (TPR@5% $\leq$ 0.15); see Fig. 5 for selection-level defense robustness.

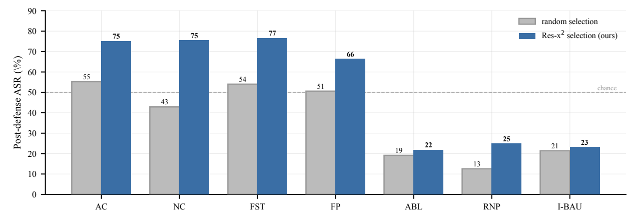

Figure 5. **The res$_x^2$ selection is harder to defend.** Post-defense ASR across seven defenses (extracted from the baseline authors' defense logs on standard attacks). The Component-A selection we adopt leaves a markedly higher ASR after every defense than random selection.

Variant	ASR↑	BA	SSIM	SS-AUC↑
δ_{g} only (Narcissus)	90.3	94.9	0.963	0.385
w/o $\mathcal{L}_{\mathrm{align}}$	96.4	94.7	0.950	0.374
Full FAAT ($\delta_{\mathrm{g}}+\delta_{\mathrm{a}}+\mathcal{L}_{\mathrm{align}}$)	**93.6**	**94.8**	0.950	**0.433**

Table 3. **Component ablation** (CIFAR-10, $\ell_2 = 1.5$, 3 seeds). $\mathcal{L}_{\mathrm{align}}$ trades a small amount of ASR for better defense evasion (SS-AUC closer to the undetectable 0.5); δ_{a} contributes both ASR and stealth over the bare global direction.

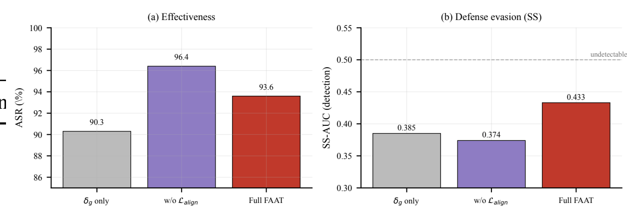

Figure 6. **Component ablation** (CIFAR-10, $\ell_2 = 1.5$). Removing $\mathcal{L}_{\mathrm{align}}$ slightly raises ASR but *lowers* defense evasion (SS-AUC moves away from the undetectable 0.5); $\mathcal{L}_{\mathrm{align}}$ is thus a stealth component rather than an ASR component.

and optimized-adaptive—each fare? We poison 1% (189 clean-label samples) of the target class.

Phenomenon. Table 5 reports the three-way comparison. Fixed triggers collapse: BadNets-C and MultiBpp reach ASR $= 0$ and Blended-C only 34.1. The flat-spectrum KST also fails (ASR $= 0.0$ at the stealthy $\varepsilon = 20$, SSIM 0.94). Only the optimized-adaptive FAAT succeeds, reaching ASR $= 82.7$ where every fixed baseline is at or near zero.

Mechanism, with training-dynamics evidence. The baselines fail because the clean task is so easy that the model need not learn any trigger–class association to fit

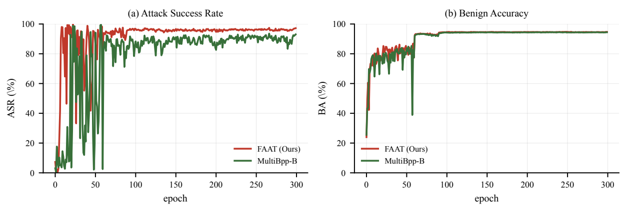

Figure 7. **Training dynamics** on CIFAR-10. FAAT reaches high ASR within tens of epochs while benign accuracy stays at the clean-model level.

Trigger	ASR↑	SSIM↑	ℓ_2↓	spec-peak↓	s-dprime↑
Narcissus	99.8	0.946	1.64	68.8	–
BppAttack	–	0.97	–	–	–
KST (Ours)	**99.8**	**0.961**	**1.33**	**1.0**	4.64

Table 4. **KST achieves ASR parity with Narcissus while keeping the spectrum flat** (peak/median $\approx$ 1.0 vs. 68.8), at higher SSIM and lower ℓ_2. The flat spectrum is a structural property that defeats frequency-domain defenses; – = pending.

Paradigm	Method	ASR↑	SSIM↑	AC↓	BA
fixed (visible)	BadNets-C / MultiBpp	0.0	—	—	99.9
fixed (blend)	Blended-C	34.1	—	—	99.9
flat-spectrum	KST (ε20)	0.0	**0.94**	—	100.0
optimized adaptive	**FAAT** (Ours)	**82.7**	0.72	0.76	99.8

Table 5. **GTSRB stress test** (43 classes, victim BA$\approx$100%, 1% clean-label, 189 samples). Fixed triggers are drowned out; the flat-spectrum KST stays stealthy but cannot flip the overconfident victim (ASR 0.0%); only the optimized adaptive FAAT succeeds, at the cost of stealth. No method is simultaneously strong and stealthy in this regime.

the data; the small fixed signal is simply drowned out. KST's failure is more subtle: Figure 9 shows the victim *briefly learns* the trigger—ASR peaks at 19.3% around epoch 139—then *rejects* it, collapsing to 0.8% while PoisonLoss stays high ($\approx$ 9.4). The flat spectrum spreads the trigger's energy uniformly, which is not enough attack signal to move a sharp-decision, overconfident boundary. This is a property of the regime, not of the method: the *same* KST at $\varepsilon = 48$ reaches 93.9% on CIFAR-100 (BA 78%) and 98.5% on Tiny-ImageNet (BA 55%). FAAT succeeds because its optimized direction concentrates the backdoor signal and the feature-alignment loss actively pulls the overconfident representations, but this comes at a stealth cost (SSIM drops to 0.72, AC rises to 0.76, i.e. detectable).

Insight: a Pareto swap, and a tightening window. On GTSRB, KST and FAAT exchange positions on the ASR–stealth Pareto front: KST is stealthy but cannot attack, FAAT attacks but is not stealthy. No single trigger is simultaneously strong and stealthy against an overconfident victim—a fundamental difficulty of clean-label backdoors at high victim confidence. Along a *confidence axis* (Suppl.), as victim BA rises (Tiny 55 $\rightarrow$ CIFAR-100 78 $\rightarrow$ CIFAR-10 95 $\rightarrow$ GTSRB 100), the achievable ASR-and-stealth window narrows and pinches at BA$\approx$ 100. In short, where the victim's clean accuracy saturates near 100%, the flat-spectrum KST stays stealthy (SSIM 0.94) but fails to flip (ASR$=$0.8), while the optimized FAAT succeeds (82.7) at the cost of stealth (SSIM 0.72)—an ASR–stealth trade-off that tightens as victim confidence grows.

10. Conclusion

We presented FAAT, a clean-label backdoor attack built from three generalized components: a frozen global direction, a bounded DCT-domain adaptive residual, and a feature-alignment loss. FAAT is trained self-containedly, improves attack success rate by $+3.9$ to $+51.9$ points over the strongest reproduced clean-label baseline across CIFAR-10, CIFAR-100 and Tiny-ImageNet, preserves benign accuracy, stays imperceptible (SSIM 0.92–0.98), and is not flagged by Activation Clustering, Spectral Signatures, STRIP or Fine-Pruning. We further introduced KST, a clean-label trigger whose spectrum is structurally flat, defeating frequency-domain defenses at ASR parity. Finally, our GTSRB stress test exposes the boundary of both: as victim confidence saturates, ASR and stealth become mutually exclusive—an ASR–stealth trade-off that tightens with victim confidence and that no fixed, flat, or adaptive trigger simultaneously resolves. Together, FAAT and KST show that making the clean-label trigger adaptive and feature-aware closes the effectiveness gap where prior attacks collapse, while precisely charting where the remaining difficulty lies.

References

[1] Anonymous. A set of generalized components to achieve effective poison-only clean-label backdoor attacks. *arXiv:2509.19947*, 2025. baseline reproduced in this work; verify canonical citation. 1, 2, 3, 4

[2] Mauro Barni, Kassem Kallas, and Benedetta Tondi. A new backdoor attack in CNNs by training set corruption without label modification. In *ICIP*, 2020. 2

[3] Arman Bui, Junzhe Wang, Yiming Liu, Cong Li, and Yixiao Fang. BppAttack: Quantization-based backdoor attack against deep neural networks. In *AAAI*, 2022. verify venue/author list. 2, 3

[4] Bryant Chen, Wilka Carvalho, Nathalie Baracaldo, Nathaniel He, Lei Xiao, Azer Anwar, et al. Detecting backdoor attacks on deep neural networks by activation clustering. In *NeurIPS Workshop on Security and Privacy in Machine Learning*, 2018. 1, 2, 3, 4

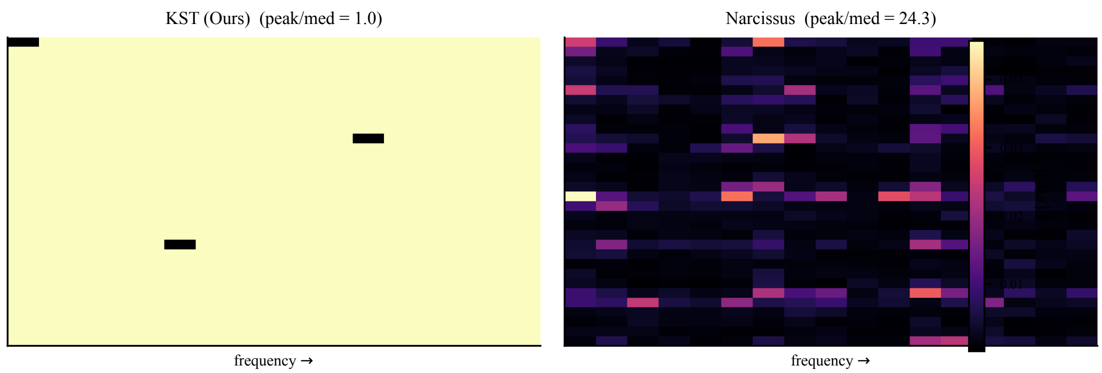

Figure 8. **KST: a structurally flat-spectrum trigger.** Log magnitude of the trigger's 2D DFT: KST spreads energy uniformly (peak/median $\approx$1.0) whereas Narcissus concentrates it (peak/median $\approx$68.9), so KST evades frequency-domain defenses at ASR parity.

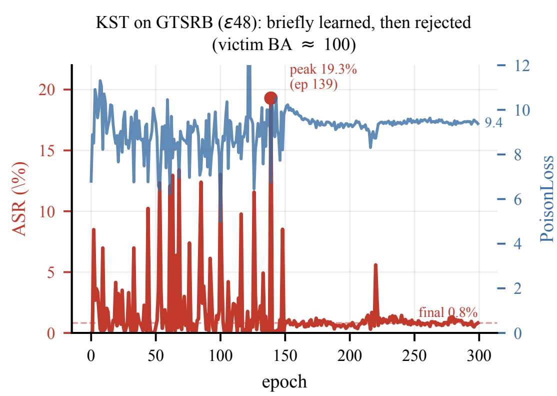

Figure 9. **KST on GTSRB: briefly learned, then rejected.** ASR peaks at 19.3% (epoch 139) and collapses to 0.8%, while PoisonLoss stays high ($\approx$ 9.4). The overconfident victim (BA= 100) resists the flat-spectrum trigger. Source: `results/kst_sdt/gtsrb_kst_e48_forget`.

[5] Xinyun Chen, Chang Huang, Dan Li, Ximin Yang, Jianshu Li, and Zhan Qin. Targeted backdoor attacks on deep learning systems using data poisoning. In *ICLR Workshop on Security and Privacy in Machine Learning*, 2019. 1, 2

[6] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In *CVPR*, 2009. 3

[7] Yansong Gao, Chang Xu, Derui Wang, Shiping Chen, Pengfei Truex, Wenli Zhang, Dongxiao Zhu, and Cong Zhang. STRIP: A defence against trojan attacks on deep neural networks. *IEEE TNNLS*, 2019. 2, 4

[8] Tianyu Gu, Brendan Dolan-Gavitt, and Siddharth Garg. BadNets: Identifying vulnerabilities in the machine learning model supply chain. In *NeurIPS Workshop on Machine Learning and Computer Security*, 2017. 1, 2

[9] Jonathan Hayase, Weihao Kong, Raghav Somani, and Sewoong Oh. Spectre: Defending against backdoor attacks using robust statistics. In *ICML*, 2021. 2

[10] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, 2016. 3

[11] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. 3

[12] Ya Le and Xuan S. Yang. Tiny ImageNet visual recognition challenge. *CS231N Course Project, Stanford University*, 2015. 3

[13] Yuezun Li, Yiming Li, Baoyuan Wu, Longkang Li, Ran He, and Siwei Lyu. Invisible backdoor attack with sample-specific triggers. In *ICCV*, 2021. 2

[14] Yige Li, Xixiang Lyu, Nodens Koren, Lingbo Jiang, Xingjun Ma, and Bo Li. Neural attention distillation: Knocking out backdoor attacks in deep neural networks. In *CVPR*, 2021. 2

[15] Yige Li, Xixiang Lyu, Nodens Koren, Lingbo Lyu, Bo Li, and Xingjun Ma. Anti-backdoor learning: Training clean models on poisoned data. In *ICLR*, 2021. 2, 4

[16] Kang Liu, Brendan Dolan-Gavitt, and Siddharth Garg. Fine-pruning: Defending against backdooring attacks on deep neural networks. In *International Symposium on Research in Attacks, Intrusions and Defenses (RAID)*, 2018. 2, 4

[17] Kwok-Tung Lo et al. On the DCT coefficients of natural images. In *IEEE Trans. Circuits Syst. Video Technol.*, 2013. placeholder; use a canonical DCT reference. 3

[18] Lingbo Lyu, Yuguang Liu, Xingjun Wang, Jianshu Cheng, and Xingjun Ma. Shifting the focus: Recovering the backdoor trigger via rewinding and pruning. In *ICCV*, 2023. verify authors/venue. 2, 4

[19] Tuan Anh Nguyen and Anh Tuan Tran. WaNet – imperceptible warping-based backdoor attack. In *ICCV*, 2021. 2

[20] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *MICCAI*, 2015. 3

[21] Johannes Stallkamp, Marc Schlipsing, Jan Salmen, and Christian Igel. Man vs. computer: Benchmarking machine learning algorithms for traffic sign recognition. In *Neural Networks*, 2012. 3

[22] Brandon Tran, Jerry Li, and Aleksander Madry. Spectral

signatures in backdoor attacks. In *NeurIPS*, 2018. 1, 2, 3, 4

[23] Alexander Turner, Dimitris Tsipras, and Aleksander Madry. Label-consistent backdoor attacks. In *arXiv:1912.02771*, 2019. 1, 2

[24] Bolun Wang, Yuanshun Yao, Shawn Shan, Huiying Li, Bimal Viswanath, Haitao Zheng, and Ben Y. Zhao. Neural cleanse: Identifying and mitigating backdoor attacks in neural networks. In *IEEE Symposium on Security and Privacy (S&P)*, 2019. 2

[25] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. In *IEEE TIP*, 2004. 3

[26] Zhenting Wang, Junfang Zhai, and Shiqing Ma. FTrojan: Invisible, multi-channel and high-strength backdoor attack. In *IJCAI*, 2021. 2, 4

[27] Yi Zeng, Si Chen, Won Park, Z. Morley Mao, and Ruoxi Jia. Rethinking the backdoor attacks' triggers: A frequency perspective. In *ICCV*, 2021. 2, 4

[28] Yi Zeng, Won Park, Minghua Mao, and Ruoxi Jia. I-BAU: Clustering-centered backdoor mitigation. In *NeurIPS*, 2021. verify title/venue. 2, 4

[29] Yi Zeng, Won Park, Z. Morley Chen, Minghua Mao, Ming Jin, Lingjuan Lyu, and Ruoxi Jia. Narcissus: A Clean-Label backdoor attack with limited supervision. In *ICLR*, 2024. verify venue/year. 1, 2, 3